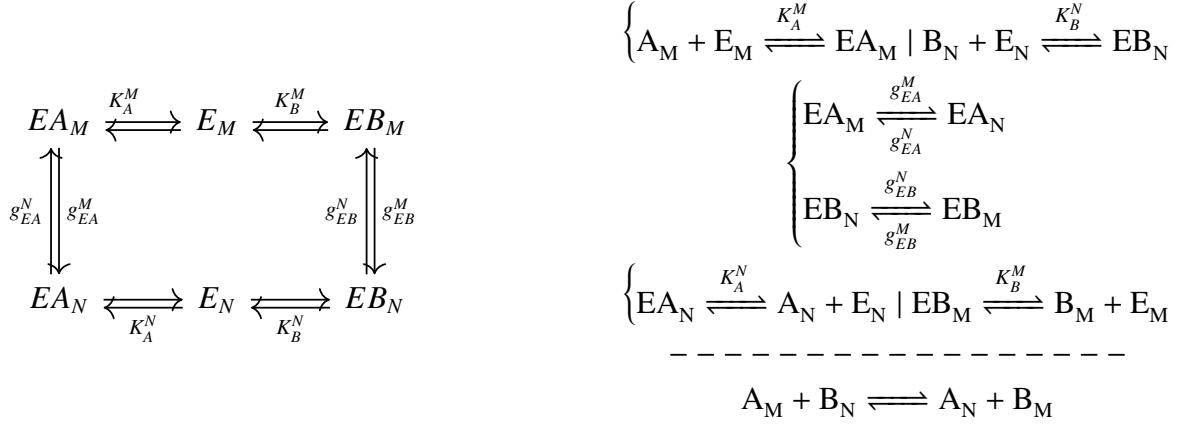


Schematic configuration of the six-state antiporter ordered binding transport mechanism



Net Transport Flux from M to N

$$J_{A,antiporter}^{M,N(net)} = [E]_t \frac{(g_{EA}^M \alpha^M)(g_{EB}^N \beta^N) - (g_{EA}^N \alpha^N)(g_{EB}^M \beta^M)}{R_M R_{NN} + R_N R_{MM}} \quad (63a)$$

$$J_{B,antiporter}^{M,N(net)} = [E]_t \frac{(g_{EB}^M \beta^M)(g_{EA}^N \alpha^N) - (g_{EB}^N \beta^N)(g_{EA}^M \alpha^M)}{R_M R_{NN} + R_N R_{MM}} \quad (63b)$$

where

$$[E]_t = [E]_M + [EA]_M + [EA]_N + [E]_N + [EB]_N + [EA]_N$$

$$\begin{aligned}
 \alpha^M &= \frac{[A]_M}{K_A^M}, & \beta^M &= \frac{[B]_M}{K_B^M} \\
 R_M &= 1 + \alpha^M + \beta^M \\
 R_{MM} &= g_{EA}^M \alpha^M + g_{EB}^M \beta^M
 \end{aligned}$$

$$\begin{aligned}
 \alpha^N &= \frac{[A]_N}{K_A^N}, & \beta^N &= \frac{[B]_N}{K_B^N} \\
 R_N &= 1 + \alpha^N + \beta^N \\
 R_{NN} &= g_{EA}^N \alpha^N + g_{EB}^N \beta^N
 \end{aligned}$$

Figure 20: A schematic configuration of the antiporter six-state mechanism of two solutes and the regarding net flux transport equation. Transport equations are reparametrized and transformed into the general resistance form from Weinstein et al. (2000) model [137], the flux equations are transformed to the general form of the resistance terms. In developing this model, it is assumed that the dissociation constants for binding reactions is not affected by binding of A and B (which means to have $K_{AB}^{MorN} = K_B^{MorN}$ and $K_{BA}^{MorN} = K_A^{MorN}$). It is also assumed that: 1) the carrier is non-symmetric and 2) binding reaction occur much faster relative to the translocation of the carrier (rapid binding assumption).

Antiporter with competitor-ordered binding model:

Figure 21 shows a competitive antiporter transport model involving molecules of A, B, and C, in which C competes with B for the same binding sites on the carrier. In this configuration, A